Tools — Shell

Tools for running shell commands and managing background tasks. `bash` commands pass through a 7-layer security model: safe-command whitelist, banned-command checks, denied patterns, directory-escape prevention, syntax validation, obfuscation detection, and user allowlist/denylist.

Tool	Description
`bash`	Execute a shell command (7-layer security).
`bash_reset`	Reset persistent shell state.
`bg`	Manage background shell tasks (spawn, list, wait, cancel).
`open`	Open a file/folder/URL via the desktop handler.

Use cases: running builds, tests, git operations, long-running tasks.

System instruction: "For simple commands use `bash` with the `command` parameter immediately. Do not describe what you will run."

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bash

Execute a shell command and return its stdout and stderr. Commands run in the agent's working directory in a persistent shell.

Arguments

Argument	Type	Required	Description	Typical value
`command`	string	yes	Shell command to execute	`"cargo test"`
`timeout`	integer	no	Timeout in seconds	`120` (default); up to `1500` for builds/tests

Example:

```
bash command="cargo test -- --nocapture" timeout=1500
bash command="cargo fmt"
```

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bash_reset

Reset the persistent shell state for the current session, clearing saved working-directory changes and environment variables. No-op if no persistent state exists.

Arguments: none.

```
bash_reset
```

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bg

Manage background shell tasks: spawn long-running commands that continue while the agent does other work, then inspect, follow, or stop them.

Arguments

Argument	Type	Required	Description	Typical value
`action`	enum	yes	`spawn`, `list`, `status`, `output`, `tail`, `cancel`, `wait`, `cleanup`	`"spawn"`
`command`	string	yes for `spawn`	Shell command to run in background	`"cargo build"`
`task_id`	string	yes for `status`, `output`, `tail`, `cancel`, `wait`	Background task id	`"bg-3"`
`lines`	integer	no	Lines returned by `tail`	`20`
`status`	enum	no	`list` filter: `running` / `completed` / `failed` / `cancelled`	—
`limit`	integer	no	Max tasks for `list`	`50`
`timeout`	integer	no	`wait` timeout in seconds	`60`
`working_dir`	string	no	Working directory for `spawn`	session cwd
`completed_only`	boolean	no	`cleanup`: only completed/failed/cancelled tasks	`true`
`older_than_minutes`	integer	no	`cleanup`: minimum task age	`60`

Example:

```
bg action="spawn" command="cargo build"
bg action="tail" task_id="bg-1" lines=20
bg action="wait" task_id="bg-1" timeout=300
```

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open

Open or reveal a file, folder, or URL using the desktop default handler (`xdg-open` on Linux, `open` on macOS, `start` on Windows). May fail in headless or sandboxed contexts.

Arguments

Argument	Type	Required	Description	Typical value
`target`	string	yes	File path, folder path, or URL	`"report.pdf"`
`action`	enum	no	`open` (default), `reveal` (open parent directory), `url` (validate and open a URL)	`"open"`

Example:

```
open target="docs/howtos/tools/pdf/file-operations.md.pdf"
```